LING10 Statistics for Linguistics
Dartmouth College. Winter 2026

Homework 3
Chi-Squared and Linear Models

Part 1: Chi-Squared

For each of the chi-squared exercises, you need to provide:
(a) A one sentence summary of whether the data set meets the assumptions of chi-squared or not.
(b) A chart (using `ggplot`) of the relationship between the two variables.
(c) A report of the results, written in formal language. You can use the examples studied in class.
(d) An explanation of the results in one or two sentences, without using any statistical words or numbers, as if you were explaining them to someone who knows nothing about statistics.

1. Warlpiri Ergatives

The dataset `warlpiri.csv` documents the use of ergative case marking in the narratives of native speakers of Lajamanu Warlpiri describing events in picture books. There are 8 children and 13 adults in the set (`AgeGroup`). The variable `CaseMarking` contains data on how often the speaker produced the ergative case versus other cases. Are the two generations different in how they use the ergative case?

The data is real: O'Shannessy, C. (2006) Language contact and child bilingual acquisition: Learning a mixed language and Warlpiri in northern Australia, PhD Thesis, University of Sydney, Australia.

2. Panamanian -s

Speakers in Panama City, like in many dialects of Spanish, delete the /s/ at the end of words:

casas 'houses'	Mexico, Costa Rica	['ka.sas]
	Panama, DR, Puerto Rico:	['ka.sah] ~ ['ka.sɑ]

Cedergren (1973) carried out a study to describe the effect of social constraints on final /s/ deletion. She recorded 8846 words whose standard Spanish variant has final /s/. For each token she recorded whether the word drops the /s/ or not (`deletion`), as well as the social class of the speakers (`class`). The data is in the dataset `panama.csv`. The structure of the variables is as follows:

`Deletion`:	whether there was a pronounced final /s/ on the word 1=deletion 0=non-deletion
`Class`:	the "social class" of the utterer 1=highest 2=second highest 3=second lowest 4=lowest

What is the relationship between social class and /s/ deletion? Your answer should have four parts:

The data is from: Cedergreen, Henrietta (1973). The interplay of social and linguistic factors in Panama. Dissertation: Cornell University.

3. Tapping in North America

In North American English, the sounds [t] and [d] can sometimes be optionally pronounced as [ɾ] (a "tap") if followed by a vowel:

For those of you who'd like to ea[t] early, lunch will be served.
For those of you who'd like to ea[ɾ] early, lunch will be served.

The dataset in `englishTap.csv` is from Kilbourn-Ceron et al. (2017), who performed an experiment to test whether speech rate affected tapping. 23 students from McGill University Participants were asked to produce sentences like these:

If you plit Alice, John will be mad.

The sentences non-words like *plit* ending in [t] or [d], followed by a vowel initial word (such as "Alice"). The speakers were asked to speak at two different rates: normal versus fast. This is in the variable `speechrate`.

How often do participants tap (`tapped`), depending on `speakingRate`?

The data is from: Kilbourn-Ceron, O., Wagner, M., and Clayards, M. (2017). The effect of production planning locality on external sandhi: A study in /t/. In Proceedings of the 52nd Annual Meeting of the Chicago Linguistic Society, pages 313–326.

Part 2: Linear Models

For each of the linear regression exercises, you need to provide:
(a) At least one chart that describes the relationship between the variables.
(b) A short discussion on whether the data meets the assumptions of a linear model.
(c) A formal report of the linear model (written as if you were submitting it to a conference or on a scientific paper).
(d) The equation that predicts the y-variable, including the coefficients, the intercept and the error.
(e) An explanation of the results in one or two sentences, without using any statistical words or numbers, as if you were explaining them to someone who knows nothing about statistics.

4. Icelandic Aspiration and Vowel Length

Icelandic consonants can be aspirated or non-aspirated, as shown below:

kapp	[ˈkʰahp]	'zeal'	*tök*	[ˈtʰœk]	'grasps' (nom pl)
gabb	[ˈkapp]	'hoax'	*dökk*	[ˈtœhk]	'dark' (fem nom sg)

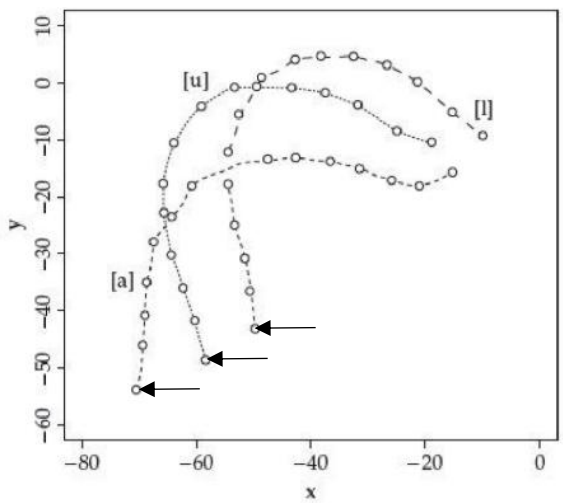
The dataset `icelandic-200.csv` has information on the duration of vowels and onsets. Does the aspiration of the onset consonant (`cons1`) change the duration of the vowel (`vowel.dur`)?

This is real data: Coretta, Stefano (2017). Vowel duration and aspiration effects in Icelandic. Masters Thesis, University of York.

5. Tongue positions that are difficult to measure

The figure below shows the shape of the tongue as people pronounce different sounds in English. These charts were produced by draping 15 small gold pellets over the tongue. However, the pellets at the back of the throat are very difficult to place. Johnson, Ladefoged & Lindau (1993) tried to figure out if there was a way to predict the x and y positions of the 15-th pellet (`x15`, `y15`) using the positions of the pellets in the front of the mouth. (The frontmost pellet is #1, the backmost pellet is #15).

The dataset `chaindata.csv` contains information collected for 54 vowels. Can the position of `y15` (the y-position of the backmost pellet) be predicted from the variables `y2`, `y6`, `y5`, `x6` and `x5`? Assume that the variables are not collinear.



This is real data: Johnson, K., Ladefoged, P., Lindau, M. (1993). *Individual Differences in Vowel Production*. J. Acoust. Soc. Am. 94(2).

6. Frequency of Dutch words

The dataset `dutch-mult-200.ccsv` has information on Dutch auxiliaries. The present perfect tense in Dutch is constructed with an auxiliary verb and a participle. There are two auxiliaries: *hebben* (as in *Ze hebben geslapen* "They slept") and *zijn* (as in *Ze zijn haar gebeld* "They called her"). Some verbs take *hebben*, some verbs take *zijn* (and some verbs take both, but we'll ignore those for now).

How can we predict the written frequency of words in Dutch? Make a linear model with `WrittenFrequency` as the dependent variable, and `FamilySize` (continuous), `LengthInLetters` (ordinal), `Auxiliary` (categorical: hebben, zijn) and `Regularity` (categorical: regular, irregular) as independent variables. Assume that the distribution of the residuals is normal and that the variance is constant.

This data is real: Baayen, R. H. and Moscoso del Prado Martin, F. (2005) *Semantic density and past-tense formation in three Germanic languages*. Language, 81, 666-698.

Evaluation

You need to turn in a Word/LibreOffice document with the answers to each of the questions. You also need to turn the R code that you used to get to your results.

Total: 100pts (12.5%)

1pt: Which you'll get for turning in the homework.

Chi-squared (15% each)
Assumptions (3pts), chart (3pts), formal report (6pts), simplified report (3pts).

Linear Models (18% each)
Assumptions (3pts), chart (3 pts), formal report (7 pts), equation (3pts), simplified report (2pts).

This homework is due at 11:59 pm of Sunday, March 1st, 2026.